Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/25/2004
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Combined Proteomic And Molecular Genetic Approaches To The Study Of Early Phases Of Root Gravitropism In Arabidopsis Thaliana		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	53706-1580	**Congressional District:**	2
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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	Gravitropism is an important process for plant growth and development. It dictates upward shoot growth and downward root growth, positioning both organs in environments that are adequate for their primary functions. Hence, long-term space-exploration missions, or colonization of other planets, will probably require that plants be engineered to better adapt to micro- and/or hypergravity. Unfortunately, this cannot be accomplished as long as we do not understand the processes that govern gravity signal transduction. Genetic and physiological strategies have identified several molecules involved in gravitropism. However, complete sequencing of the Arabidopsis thaliana genome revealed that most genes are members of gene families, suggesting that widespread functional redundancy might complicate forward genetics. There is a need for alternative approaches to the identification of new gravity signal transducers. Using two-dimensional gel electrophoresis, we identified root-tip proteins whose abundance, modification and/or cellular localization are		

Task Description:	affected early after gravistimulation. Several of them were affected within the first 12 minutes of stimulation, preceding the curvature response. Among these, an unusual response regulator (ARR22) was up-regulated during this period. Typically, response regulators are implicated in signal transduction pathways that involve two-component receptors. This proposal will describe experiments aimed at identifying more of these proteins by mass spectrometry, and investigating their distribution in root-tip cells before and after gravistimulation. Mutations will be isolated in the corresponding genes (including ARR22), and their phenotypes will be analyzed. If/when a gravitropic defect is found, we will collaborate with other investigators to analyze the effect of these mutations on amyloplast sedimentation, the visco-elastic properties of the statocytes, the gravity-induced relocalization of auxin efflux carriers in the root tip, and cytoplasmic alkalinization of the statocytes early after gravistimulation. This multidisciplinary approach will fill some of the gaps that hinder our understanding of gravity perception and signal transduction in plants. Objective of this task is to identify and characterize root-tip proteins that are differentially expressed, modified or targeted early in response to gravity stimulation. While Genetics has been fruitful at identifying genes that contribute to gravity signal transduction in the root tip of plants, it also suffers from limitations due to functional redundancy. Proteomic strategies allow direct access to proteins whose abundance, localization and/or modification vary in response to gravity stimulation, regardless of gene redundancy. When such proteins are found, reverse genetics can then be used to characterize their involvement in gravity signal transduction. This proteomic approach complements microarray analyses of gene-expression responses to gravity stimulation because they can account for translational and post-translational modifications, which constitute potential regulatory mechanisms in gravity signal transduction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will allow us to better understand the molecular mechanisms that govern gravity signal transduction in plants. In turn, it may provide the knowledge and tools that are needed to engineer plants that display a better ability to respond to lodging by wind and rain, a major source of crop loss in the world. Furthermore, the new proteomic approaches that are being developed as a part of this project are likely to be very useful in the study of multiple developmental processes and responses to environmental stimuli in plants.
Task Progress:	Using two-dimensional (2-D) gel electrophoresis coupled with mass-spectrometric identification of proteins, we have identified 30 new root-tip proteins as being differentially expressed, modified and/or targeted in response to gravity stimulation in Arabidopsis thaliana seedlings. We grouped these proteins into functional classes: Metabolism (20), transcription factors (2), kinase (1), cytoskeleton-related (1), chaperone (1), and unknown (5). Amongst these proteins, only two were also differentially expressed in response to the mechano-stimulus that accompanies gravity stimulation in our assays, indicating that the other 28 proteins are responding specifically to gravity stimulation. To overcome the tedious nature of experiments involving 2-D gel electrophoresis, we also worked on the development of a higher throughput approach, based on SDS-polyacrylamide gel electrophoresis coupled to liquid chromatography and tandem mass spectrometry, in collaboration with the Mass Spec Facility at the University of Wisconsin – Madison. Using this approach, we identified approximately 1,500 different proteins in our extracts, and obtained preliminary evidence for differential representation of several of these proteins between stimulated and control samples. Although this approach remains purely qualitative, we are working on strategies aimed at improving their quantitative reliability. Three of the differentially represented proteins identified by 2-D gel electrophoresis turned out to be enzymes involved in the AdoMet cycle, an important metabolic pathway that provides precursors for transmethylation reactions and for the biosynthesis of plant growth regulators such as ethylene, polyamines, and nucleotides. We continued our analysis of adenosine kinase (ADK), an enzyme that salvages a by-product, negative-regulator of that pathway (Adenosine). We have shown that this protein functions in gravity signal transduction within Arabidopsis roots tips, as well as in root-cap morphogenesis and auxin accumulation within the columella region of the cap. Current research is aimed at defining how ADK functions in these pathways. Overall, this analysis of ADK's involvement in root-tip gravitropism provided a nice proof-of-concept for our proteomic approach, and led us to identify new mechanisms in the control of gravity signal transduction within Arabidopsis root tips.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Articles in Peer-reviewed Journals	Blancaflor, E.B.; Masson, P.H. "Plant gravitropism. Unraveling the ups and downs of a complex process." Plant Physiology 133(4):1677-1690, Dec 2003. , Dec-2003
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Presentation	Boonsirichai, K.; Harrison, B.; Stanga, J.; Young, L.-S.; Neal, C.; Sabat, G.; Murthy, N.; Harms, A.; Sebrook, J.; Masson, P. "Genetic analysis of gravity signal transduction in Arabidopsis thaliana seedlings." 35th COSPAR Scientific Assembly, Session F1.1. Jul-2004
Presentation	Masson, P.H. "Molecular genetic analysis of root gravitropism in Arabidopsis thaliana." Seminar, Max-Planck Institute, Cologne, Germany. May-2004
Presentation	Masson, P.H. "Proteins involved in gravity signal transduction in Arabidopsis root tips." Seminar, University of Freiburg, Germany. Jul-2004
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